TOPICAL TREATMENT OVERVIEW for Atopic Dermatitis, Psoriasis, Seborrheic Dermatitis

M Jin PharmD © www.RxFiles.ca Sept 2026

	Atopic Dermatitis (AD) aka Eczema [1-4,53-56]	Plaque Psoriasis (PP) [5-7]	Seborrheic Dermatitis (SebD) [8]
Clinical Presentation	• Dry or crusting, flaky, itchy skin, "*the itch that rashes*" • Usually begins in childhood, often improves in adulthood • Chronic inflammatory skin disorder frequently associated with food allergies, allergic rhinitis, and asthma.	• Erythematous lesions with silvery-white scales, usually itchy. • Auspitz sign (tiny, pinpoint bleeding under scales) • Any age, typically peaks between 20-30yr and 50-60yr • Chronic condition; symptoms may resolve/recur throughout life	• Greasy yellowish scale at sebum sites (e.g. cheeks, scalp, nasolabial folds, eyebrows, beards, ears), minimal itch • Infancy (cradle cap) & adults 30-60yr • Chronic, relapsing, noninfectious condition
Skin of Colour Considerations	• In darker skin, may present as dry, flaky, excoriated patches with post-inflammatory hypopigmentation. • Favouring cheeks and scalp in infants and flexural areas in older individuals	• Erythema (redness) may be subtle; plaques may be thicker. • Untreated psoriasis can lead to more dyspigmentation if not treated early. Consider early initiation of systemic tx.	• In darker skin, hypopigmented, mildly scaly patches in the eyebrows, glabella, & nasolabial folds are common • For scalp, use OTC leave-in preparations[spray/serum] with zinc pyrithione **Head & Shoulders**, or TCS oils/ointments
	• Topical corticosteroids (TCS) can cause dyspigmentation or hypopigmentation, calcineurin inhibitors (CNI) or other non-corticosteroid topicals may be preferred • Phototherapy can darken skin • For the scalp, coiled hair types are more susceptible to dryness and breakage which can be exacerbated by alcohol in topical gels/solutions/shampoos. Foams or lotions may be preferred. (e.g. Clobetasol propionate 0.05% spray **CLOBEX**, betamethasone valerate 0.1% scalp lotion **BETADERM**/**ECTOSONE**, calcipotriol/betamethasone dipropionate foam **ENSTILAR**, roflumilast foam **ZORYVE**)		
Pharmacological Agents: See RxFiles Charts: Topical Corticosteroids pg 32, Antifungals pg 121, **Topical NON-Corticosteroids** (next pg), and **Table 1 Table 2** below. Consider efficacy, cost, pt preference & convenience when selecting drug(s) & formulation.			
Approach to Therapy			
1st line Non-pharm (Phototherapy, if available)	**Use non-medicated moisturizers -** use daily & long-term to prevent flares in atopic/seborrheic dermatitis, psoriasis or dry skin. One moisturizer type unlikely better than another.[9] See **Table 1** below.		
	• Avoid contact with irritants/trigger factors (e.g. wool, nylon) • Wash clothing in soap vs detergent; double rinse/vinegar • Limit use of soap to axillae, feet, and groin[2]	• **Lifestyle:** Quit smoking, limit alcohol, exercise, diet[Mediterranean], ↓ wt if obese, ↓ stress (e.g. meditation, mindfulness, CBT), acupuncture[10,11] • **Aloe Vera, curcumin, Indigo Naturalis, St. John's Wort, Omega-3 fish oil**	• **Mineral or coconut oil** for thick scaling • **Cradle cap is benign and self-limiting** – use non-medicated shampoo frequently and gently brush scalp
	Wet wrap therapy or occlusive dressings to emollient or TCS may help for ↑ penetration, ↓ water loss, & act as a physical barrier against scratching for moderate-severe atopic dermatitis or psoriasis. Use short term (7-14 d). Whole-body dressing or wet wrap with other drugs (e.g. CNI) should be initiated by specialist.[2]		**Wet wrap tx** is not recommended in SebD due to possible overgrowth of *Malassezia* yeast.
Topical Medicated Drug Choices	➢ If symptoms (e.g. itch) are not controlled with non-pharm, recommend **topical corticosteroids (TCS) once-twice daily until resolved (~2-4wk)**, then PRN (continue daily moisturizers). Determine TCS potency based on the site/severity {e.g. low potency TCS {e.g. hydrocortisone 0.5%} for face, folds, intertriginous areas (armpits, groin, under breasts, genitals, buttocks), large areas of body, or mild severity}. Plaque psoriasis may require higher potency TCS depending on the site (see **Table 2** below). If intolerable AEs or lack of efficacy, assess adherence & proper amount applied before considering a different TCS potency/formulation or add/switch to another drug. See RxFiles Chart: Topical Corticosteroids pg 32 If TCS is not effective, the next treatment option varies based on site, severity, indication, pt preference and cost. *There is no single ideal combination or sequence of agents for AD or PP*, but *tx should be kept as simple as possible*. Consider the following alternatives for AD or PP: • Calcineurin inhibitors (CNI) – **tacrolimus PROTOPIC** or **pimecrolimus ELIDEL** for sensitive areas (e.g. face, intertriginous, genitals). Considered 2nd line for atopic dermatitis; off-label for plaque psoriasis or seborrheic dermatitis. • PDE4-Θ – **roflumilast ZORYVE** crm/foam is indicated in AD, PP, & SebD, but is more expensive - $350/60g. Effective vs placebo.[36,37] • Aryl hydrocarbon receptor (AhR) agonist – **tapinarof NDUVRA** is used in AD and PP, but more expensive, $500/60g. • Combination of TCS with non-TCS topicals (e.g. CNI, VDA, retinoid) are considered more effective than monotherapy. Once under control, can ↓ amount (e.g. ① TCS on weekends + non-TCS on weekdays; ② TCS in am + non-TCS HS; or ③ non-TCS 2-3x/wk to areas that flare). Some combination formulations include: **DOVOBET**, **ENSTILAR**, **WYNZORA**, **DUOBRII**, **DIPROSALIC**. See next pg for more info.	• SebD can be treated with topical antifungal(s) &/or TCS. • For acute flares or inflammation, consider TCS (e.g. hydrocortisone for the face or mometasone 0.1% for the body) for a short period (e.g. 1-2 wk), then transition to less frequent use for long-term management. • Consider topical antifungals (e.g. **ciclopirox** 1% **LOPROX**, **ketoconazole KETODERM**) if TCS is not effective. • For scalp, consider medicated shampoos (e.g. **ketoconazole NIZORAL**, **selenium sulfide SELSUN BLUE**, **zinc pyrithione HEAD & SHOULDERS**, **salicylic acid DERMAREST**, sulfur-based, **coal tar**) • If topical antifungals or TCS are not effective, consider CNI {off-label} – **tacrolimus PROTOPIC**, **pimecrolimus ELIDEL** or PDE4-Θ – **roflumilast ZORYVE** (more expensive).
	Other alternatives for atopic dermatitis only: • JAK-Θ – **ruxolitinib OPZELURA**, **delgocitinib ANZUPGO** (hand) • PDE4-Θ – **crisaborole EUCRISA** for atopic dermatitis age≥**3mo** **Avoid** oral corticosteroids (rebound flares in AD) or oral antihistamines (AD is not histamine related).[14,57]	Other topical alternatives for plaque psoriasis only: • Vitamin D Analogues (VDA) – **calcipotriol DOVONEX**, **calcitriol SILKIS** are effective as TCS with potentially more skin irritation.[39,40] • Retinoid – **tazarotene ARAZLO** best for indurated lesions • **Anthralin** 0.1% compounded if other drug(s) are ineffective • **OTC: Coal Tar**, **Salicylic Acid** if cost is an issue	Due to the chronicity of atopic dermatitis, plaque psoriasis or sebhorrhic dermatitis, the **duration of therapy** for topical medication varies depending on the severity/location. If symptoms have not improved after 4 wk, re-evaluate the differential diagnosis &/or adherence to topicals before changing to another topical tx &/or refer to specialist.
Systemic drugs	Refer to specialist for systemic drugs (e.g. azathioprine, oral JAK-Θ, cyclosporine, methotrexate, mycophenolate mofetil, or biologic)	Refer to specialist for systemic drugs (e.g. acitretin, apremilast, cyclosporine, deucravacitinib, methotrexate, or biologic)	• Refer to specialist for systemic drugs (e.g. fluconazole, isotretinoin, itraconazole, or terbinafine)
Pregnancy: **Avoid retinoids**, AhR agonist, JAK-Θ, roflumilast, salicylic acid; use lowest potency TCS with shortest duration. **Lactation:** **Avoid** high TCS, retinoid, ruxolitinib, &/or medicated topicals around the nipple. Limit/avoid sun & UV light exposure with: coal tar, calcineurin inhibitors, JAK-Θ, St. John's Wort, retinoids, vitamin D analogs. Risk of photosensitivity, phototoxicity, or skin cancer. Use appropriate sun protection.			

TABLE 1: COMMON MOISTURIZERS, SALICYLIC ACID, COAL TAR AND OTHER OTC PRODUCTS FOR ATOPIC DERMATITIS (ECZEMA), PSORIASIS OR SEBORRHEIC DERMATITIS

MOISTURIZERS: Apply daily-BID after bathing within 3-5 min.[9] Sunflower/coconut oil[Extra Virgin or Cold Pressed Coconut Oil] may be beneficial for eczema.[19] If sensitive skin, **avoid** products with fragrance, lanolin, bath salts. See **OTC** examples (prices $ vary):
1. Emollients (such as colloidal oatmeal, glycol, glyceryl stearate, shea butter, soy sterols): **AVEENO**, **DORMER 211** Hyper Emollient, **GLAXAL BASE**, **ALPHA KERI**, **LUBRIDERM**, **NEUTROGENA**
2. Humectants/Hydrating (e.g. alpha-hydroxy acids, glycerin, phospholipids, propylene glycol, urea): **DERMAL THERAPY**, **GLYCERIN**/**GLYSOMED**, **NEOSTRATA**, **UREMOL**/**URISEC** ✗ ▼ [urea 10%, 20%, 22%, 40% crm; 10-12% lot]
3. Barrier Repairing Agents (ceramides): **CERAVE**, **CETAPHIL** **4.Occlusives** (such as petrolatum, dimethicone, mineral oil): **EUCERIN**, **NUTRADERM**, **VASELINE**
5. Bath products: **ALPHA KERI** bath oil/bar, **AVEENO** bath oil/bath powder/bar, **CERAVE** cleanser/wash/shampoo/bar, **CETAPHIL** cleanser/wash/oil
Refer to Skincare products or Product Recognition Programs endorsed by Canadian Dermatology Association & Seal of Acceptance by Eczema Society of Canada

SALICYLIC ACID (SA) 2%-5%: PL Produces desquamation of hyperkeratotic epithelium; useful in removing scales in psoriasis/seborrheic dermatitis. ✓Plaque psoriasis, hyperkeratosis of palms/soles.
Combining SA with TCS or tacrolimus ↑ drug penetration & efficacy for plaque psoriasis.[20] **CAUTION**: renal/hepatic impairment, age ≤18yr with viral illness due to risk of Reye's syndrome with salicylates.
AE: erythema, scaling, local irritation, rare-salicylate toxicity (prolonged use over large areas > 20% BSA, especially in children). **DI**: salicylic acid ↓ calcipotriol efficacy - use 10-12hr apart.
OTC: Shampoo: SA 4% **SEBCUR** ▼ $12[120mL], SA 3% **DERMAREST** $10[236mL], **SELSUN BLUE BOTANICAL** $10[300mL]; **Face wash cleanser**: **GARNIER PURIFYING GENTLE CLEANSER** $15[240mL]

COAL TAR (CT): PL Anti-inflammatory, anti-pruritic activity. ✓Relief of symptoms (itching, scaling, flaking, redness, and irritation) due to dandruff, seborrheic dermatitis, or psoriasis. Potentially steroid sparing.
AE: folliculitis, contact dermatitis, phototoxicity, photosensitivity, stain skin/clothes/sinks/tubs, odor, possible carcinogenicity (controversial).[21,22]
OTC: Shampoo: Coal tar 0.5% **COAL-TAR** $23[240mL]. **Solution:** **Odan-COAL TAR** ✗ ▼ 20% solution $22[100mL]. **COMBINATION SA+CT**: SA 4% + coal tar 10% **SEBCUR T** ✗ ▼ $16[120mL], $25[240mL]; SA 2% + coal tar 5% **TARDAN** $23[250mL]

OTHER **OTC: Indigo Naturalis** topical herbal balm/salve/ointment, **aloe vera** (**CI**: if allergy to lily family/onion/garlic), topical **St. John's Wort** (can cause photosensitivity, avoid concomitant phototherapy), **Omega-3 fish oil** (oral).[10,11]
Astringent (aluminum sulfate tetradecahydrate, calcium acetate monohydrate) topical solution **DERMBURO** for minor skin irritations, eczema, poison ivy, insect bites, athlete's foot, & contact dermatitis; $20/12x3g sachet.

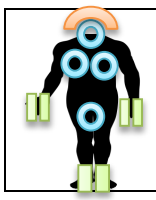


TABLE 2: LOCATION & TOPICAL PREFERENCE – consider pt preference, convenience, adherence, availability, & cost. ONE topical agent for multiple sites may be preferred to ↓ confusion.

Location	Preferred vehicle	Preferred topical examples in psoriasis[expert]	Comments
Scalp	Cream, foam, gel, lotion	**Clobetasol propionate** 0.05% spray **CLOBEX** (least expensive), **roflumilast** 0.5% cream/foam **ZORYVE**	Ointments are not ideal cosmetically
Face, intertriginous areas, genitals, folds	Cream, foam, gel	**Hydrocortisone valerate** 0.2% cream **HYDROVAL** (least expensive), **roflumilast** 0.5% cream **ZORYVE**, CNI - **pimecrolimus** 1% cream **ELIDEL**, **tacrolimus** 0.03% or 0.1% ointment **PROTOPIC**	**Avoid** high-mid potency TCS (Group 1-5) and VDA (**calcipotriol**, **calcitriol**)
Body	Cream, foam, lotion, ointment	Combination products: **WYNZORA** cream, **DOVOBET** foam/ointment, **ENSTILAR FOAM**, **DUOBRII** lotion As per monograph, restrict **WYNZORA** cream to BSA ≤ 30% per application	**Avoid** high-potency TCS (Groups 1-3) if BSA > 20%. Restrict **ruxolitinib OPZELURA** to BSA ≤ 20% per application
Palms or Soles	Cream, foam, ointment	**Clobetasol propionate** 0.05% crm/ointment **DERMOVATE**, **roflumilast** 0.5% crm/foam **ZORYVE**, **ENSTILAR FOAM**	May require more occlusion, VDA, retinoid, AhR, SA/CT

TOPICAL NON-CORTICOSTEROID FOR ATOPIC DERMATITIS (ECZEMA), PSORIASIS, & SEBORRHEIC DERMATITIS: Comparison Chart

M Jin PharmD © www.RxFiles.ca Sept 2026

Generic/TRADE {Alphabetical order by drug class}	Indication, Efficacy	Application site (See Table 2)	Adverse Events AE / Contraindications CI / Drug Interactions DI / Monitoring M / Comments	Apply (MAX)	$/pkg
ARYL HYDROCARBON RECEPTOR (AhR) AGONIST: binds to and activates AhR, a ligand-dependent transcription factor that regulates cytokine, skin-barrier expression and antioxidant activity. Exact MOA unknown.					
Tapinarof NDUVRA 1% cream ✗ ⊗ (USA: **VTAMA**) PL	CAN only: ✓Plaque psoriasis ≥18yr US FDA: ✓ Atopic dermatitis ≥2yr & plaque psoriasis	All body areas	Odourless non-staining cream. Avoid swimming, bathing, showering, or strenuous activities for at least 2 hours after application. AE: Local: folliculitis[up to 20%], contact dermatitis[7%], pruritus[3%], urticaria[<1%]. Systemic: HA[4%], influenza[2%], nasopharyngitis[11%], URTI[12%].	Once daily	$500[60g]
CALCINEURIN INHIBITORS (CNI): binds to the intracellular protein FKBP-12, inhibiting calcineurin, which blocks cytokine transcription and inhibits T-cell activation. Not to be used with occlusive dressings.					
Pimecrolimus (PIM) ELIDEL 1% cream ☎▼ PL	✓Atopic dermatitis (AD) ≥2yr (off-label psoriasis, seborrheic dermatitis, vitiligo). For AD, TAC more effective than PIM over 6wk.[24] PIM less effective than moderately potent topical corticosteroids (TCS); TAC more effective than mild TCS and as effective as moderately potent TCS.[25]	✓ consider for thinner skin (e.g. face, intertriginous areas).	AE: Local: burn/itch[up to 60%] ↓ over time *(in psoriasis, apply TCS x 2-3 days before CNI, refrigerate 15-20 min prior to application, &/or apply on dry skin to minimize burning/irritation)[expert]*, erythema[28%], acne[4%], folliculitis[6%], skin infection[12%], urticaria[6%], sunburn[2%]- **Avoid** sun, **skin cancer** < 0.1%.[26-31] Systemic: flu-like symptoms[31%], cough[18%], fever, & HA esp in **kids**[25%], flushing post-alcohol, GI[1-8%] (pain, constipation, D/N/V, gastroenteritis)	BID	$110[30g], $210[60g]
Tacrolimus (TAC) PROTOPIC 0.03%, 0.1% ointment ☎▼ PL				BID until clear; once daily 2x/wk 2-15yr: 0.03%	0.03%, $237[60g] 0.1%, $253[60g]
JANUS KINASE INHIBITORS (JAK-Θ): inhibits JAK1 and JAK2 which mediate signaling of cytokine factors responsible for inflammation, itch response and skin barrier function. **Immunizations:** Complete all age-appropriate immunizations before initiating therapy; avoid administration of **live vaccines** immediately prior to, during, and immediately after therapy. Vaccinate for shingles prior to starting JAK-Θ.					
Delgocitinib ANZUPGO 20mg/g cream ✗ ⊗ PL	✓Moderate to severe chronic hand eczema in adults ≥ 18 yr.	Hand eczema	AE (≤1%): Local: skin infection, erythema, pain, paronychia, pruritus. Systemic: herpes zoster, leuko-/neutro-penia, paresthesia, skin carcinoma[non-melanoma].	BID (60g/mo)	$720[60g]
Ruxolitinib OPZELURA 1.5% cream ✗ ⊗ PL	✓Atopic dermatitis & nonsegmental vitiligo ≥12yr. US Boxed Warnings for all JAK-Θ systemic and topicals: serious infection, mortality, malignancies/lymphoma, MACE, thrombosis.	All body areas, restrict to ≤20% BSA per application	AE: Local: acne[6%], dermatitis[<1%], erythema[2%], (irritation, pain, pruritus)[≤5%], urticaria[1%]. Systemic: fever[≤2%], HA[4%], herpes zoster, nasopharyngitis[4%], URTI[>10%], UTI[2%] **Serious (rare)**: cytopenia, infection, nonmelanoma skin carcinoma. D/C ruxolitinib in serious infection, hx of MI/stroke, thrombosis (PE/DVT) or skin cancer.	BID (60g/wk or 100g/2wk)	$1150[100g]
PHOSPHODIESTERASE 4 INHIBITORS (PDE4-Θ): causes accumulation of intracellular cyclic AMP (cAMP) and reduced expression of pro-inflammatory cytokines associated with psoriasis.					
Crisaborole EUCRISA 2% ointment ✗ ⊗ PL	✓Atopic dermatitis age≥3mo. Effective vs placebo, no head-to-head RCTs vs other topicals.[31-33]	All body areas	AE: Local: Application site (pain, burning, stinging)[4%], urticaria[≤1%].	Once daily - BID	$180[60g]
Roflumilast ZORYVE 0.05% USA, 0.15%, 0.3% cream; 0.3% foam ✗ ⊗ PL	✓Atopic dermatitis 0.15% crm ≥6yr & 0.05% age 2-5y ✓Plaque psoriasis 0.3% crm/foam ≥12yr ✓Seborrheic dermatitis 0.3% foam ≥9yr Effective vs placebo.[36,37]		AE: Topical: pain[1-2%]; Systemic: diarrhea[≤3%], HA[3%], insomnia[≤1%], rhinitis[2%], URTI[4%].[35] CI: Child-Pugh B/C liver dx; DI: CYP3A4 inhibitors, OC (gestodene, ethinyl estradiol)	Once daily	$350[60g]
RETINOID: Synthetic, acetylenic retinoid which modulates differentiation and proliferation of epithelial tissue and exerts some degree of anti-inflammatory and immunological activity. See RxFiles Chart: Acne, pg 33					
Tazarotene (TAZ) ARAZLO 0.045% lotion (USA: **TAZORAC** 0.05%, 0.1% gel & cream available) PL	✓Acne (Off-label psoriasis). In psoriasis, TAZ gel has ↓ success rate for all lesions compared to fluocinonide 0.05% crm BID.[38] D/C once control is achieved, therapeutic effect may persist after tx is stopped.	All body areas	AE: Local: (>10%) Application site pain, burning, pruritis, erythema, dry skin, contact dermatitis, dyspigmentation, photosensitivity (↑ risk sunburn). Has slightly higher incidence of AE than TCS or VDA. Systemic: facial pain, peripheral edema, ↑TG. CI: Pregnancy - birth defect risk	Once daily (HS)	$80[45g]
VITAMIN D ANALOGUES (VDA): regulates skin cell production and proliferation. Binds to vitamin D receptors and inhibits keratinocyte proliferation and enhances keratinocyte differentiation.					
Calcipotriol DOVONEX 50mcg/g ointment (=**calcipotriene** in USA: **SORILUX** foam, **VECTICAL/CALCITRENE** ointment) PL	✓Plaque psoriasis ≥2yr. VDA as effective as TCS but more skin irritation.[39,40] UVA exposure can inactivate; apply at HS.[39]	**CI: Avoid** face, intertriginous (folds) areas	AE: Local: (burning, stinging, tingling)[≤23%], pruritus[15%], rash[≤11%], pain[3%], (dryness, erythema, peeling)[≤10%], photosensitivity, urticaria (rare). Systemic: Hypercalcemia[<1%], hypercalciuria[rare] M: Calcium DI: **Caution meds which ↑ Ca^{++}** (e.g. oral Vitamin D, calcium or thiazide diuretics); acidic topicals (e.g. SA, hydrocortisone valerate) ↓ calcipotriol efficacy - use 10-12 hr apart.	Daily-BID (HS) (Max: 2-5yr: 25g/wk 6-10yr: 50g/wk 11-14yr: 75g/wk >14yr: 100g/wk)	$155[120g]
Calcitriol SILKIS ✗ ⊗ 3mcg/g ointment PL	✓Mild to moderate plaque psoriasis ≥18yr. Calcitriol is better tolerated than calcipotriol.[41-44]		AE: Local: pain[3%]; rare: burning, erythema, bullous dermatitis (acute) Systemic: Hypercalcemia[24%] M: Calcium DI: **Caution meds which ↑ Ca^{++}** (e.g. oral Vitamin D, calcium or thiazide diuretics)	BID (Max: 2-6yr: 100g/wk ≥7yr: 200g/wk)	$110[60g]
COMBINATIONS: Combo strategies of above for psoriasis: ① VDA weekdays & TCS weekends or ② TCS am & VDA HS. See ingredients above for AE, CI, DI, M. In general, combination therapy more effective than monotherapy.[45]					
Calcipotriol + Betamethasone dipropionate **DOVOBET g; ENSTILAR** foam, **WYNZORA** crm 50mcg/0.5mg ointment, gel, foam, crm (USA: **TACLONEX** ointment/susp) PL	**DOVOBET, g**: ✓Psoriasis vulgaris ≥ 18 yr **ENSTILAR**:foam, **WYNZORA** crm: ✓Psoriasis vulgaris ≥12yr Cream: **total BSA should not exceed 30%**	**Avoid** face, intertriginous (folds), genital areas	Combination VDA + TCS is more effective than VDA alone in psoriasis; can result in ↓ AE & thinning of the skin.[46-49] Improvement as early as 1-2 wk. After control is achieved with initial tx, maintenance application 2-3 times/wk may be helpful in preventing/limiting future flare.[16] It is recommended to allow 8 hr between application & showering to avoid washing it off.	Once daily **Crm, Oint., Gel:** (100g/wk) **Foam:** (60g q4d) **Crm:** (15g/d)	**Ointment:** $100 g[60g], $200[80g] $150 g, $245[120g] **Gel:** $100 g[60g], $145[80g] **Foam:** $120[60g] **Cream:** $120[60g]
Tazarotene + Halobetasol DUOBRII 0.045%/0.01% lotion PL	✓Plaque psoriasis (moderate-severe) in adults ≥ 18 yr		Combination tazarotene + halobetasol supports synergy in moderate-severe psoriasis more effectively than monotherapy, ↑ total remission time, prevents thinning of the skin, & less skin irritation.[50-52]	Once daily (50g/wk)	$220[100g]
Salicylic acid + Betamethasone dipropionate DIPROSALIC 3%/0.05% ointment PL	✓Dermatoses in ≥18yr		**Avoid** under occlusive dressing. Safety in pediatric patients has not been established.	BID	$75[50g]
OTHER: Anthralin 0.1% compounded similarly effective as calcipotriol for psoriasis. **Avoid** face or other sensitive areas. AE: burning, perilesional erythema, skin staining.					

Irritation/tolerability may depend on individual, site/location, indication, & amount. The following is an estimate of the LEAST irritating to the MOST irritating topicals:

LEAST irritating → ① Non-medicated moisturizers (e.g. petrolatum) → ② TCS → ③ tapinarof → ④ PDE4-Θ (roflumilast < crisaborole) → ⑤ CNI (pimecrolimus < tacrolimus) → ⑥ VDA (calcipotriol < calcitriol) → ⑦ JAK-Θ (delgocitinib < ruxolitinib) → ⑧ SA → ⑨ CT → ⑩ tazarotene → **MOST irritating**

SYSTEMIC ORAL AGENTS for Atopic Dermatitis (AD) and Psoriasis: Drug Comparison Chart[55-60]

M Jin PharmD, D Parekh PharmD © www.RxFiles.ca Sept 2026

Generic/TRADE	Indication, Efficacy, Place in Therapy	Adverse Events AE / Contraindications CI / Drug Interactions DI / Monitoring M Comments	Dosing (MAX)	$/30 day
PHOSPHODIESTERASE INHIBITOR (PDE4-Θ): Anti-inflammatory				
Apremilast, OTEZLA, g ✗ ⊗ 10mg, 20mg, 30mg tablets	✓Moderate-severe psoriasis	AE: **Common** (>10%): diarrhea, nausea, HA, URTI, weight loss (↓ 5-10%). Other: ↓appetite[3%], abdominal discomfort/pain[9%], vomiting[9%], fatigue[3%], insomnia[2%], arthralgia[6%], back pain[8%], depression[≤1%], suicidal ideation[<1%]. DI: ↓apremilast: St John's Wort, CYP3A4 inducers[rifampin, phenobarbital, carbamazepine, phenytoin] M: wt, renal fx, mood changes, depression, suicidal ideation. CI: pregnancy	Start 10mg po daily, ↑10mg/d until 30mg BID by day 6 CrCl <30ml/min: 30mg/d	14 day starter kit: $570, g 30mg BID: $1200, g
RETINOID: binds to & activates retinoid X/acid receptors to inhibit the proinflammatory cytokines interleukin-6, migration inhibitory factor-related protein-8, & interferon gamma resuilting in anti-inflammatory & antiproliferative actions to the epithelium.				
Acitretin SORIATANE, g ☎ ▼ 10mg, 25mg capsules Hazardous drug handling precautions	✓Severe psoriasis Takes 3-6mo for maximal response. Efficacy is dose-dependent. Rarely used as monotherapy, typically combined with TCS, ustekinumab, or phototherapy.[61]	AE: **Common**: alopecia, ↑LFT (AST), dry mouth/skin/eyes, HA, insomnia, joint pain, GI, ↑TGs, depression, tinnitus/impaired hearing, photosensitivity, epistaxis, brittle nails, chelitis. **Rare**: pancreatitis, pseudotumor cerebri, ↓ night/color vision, impaired glucose CI: pregnancy, severe renal/liver dysfunction, chronic hyperlipidemia DI: CYP P450, alcohol (creates etretinate that can ↑ & prolong teratogenic effects), MTX (↑ hepatotoxicity), tetracyclines (↑ intracranial pressure-pseudotumor cerebri, Progestin minipill (↓effectiveness of progestin); vitamin A supplementation M: Baseline: pregnancy test(s), CBC, lipids, LFTs, SCr, vision, BG in DM, BMD in elderly. Ongoing: pregnancy monthly during tx & q3mo for ≥ 3yrs after D/C acitretin; LFTs q4wk for 1st 3 mo, then q3mo; CBC & SCr q3mo Blood donation not recommended while on and for 3 yr after due to teratogenicity.	10-**35**-75mg/day 35mg/day appears to be optimal for efficacy/safety (Max: 75mg/d)	$50-130-235
IMMUNOSUPPRESSANTS: Anti-inflammatory. See RxFiles Chart: Inflammatory Bowel Disease Overview pg 75, Rheumatoid Arthritis pg 145, Solid Organ Transplantation pg 229 for more AE, DI, CI, M. **Azathioprine** (AZA), **cyclosporine** (CSA), **methotrexate** (MTX), and **mycophenolate mofetil** (MMF) are **off-label** for moderate-severe atopic dermatitis or psoriasis & usually prescribed by specialist.				
Cyclosporine (CSA) NEORAL, g [g ✗ ▼] 10, 25, 50, 100mg cap; 100mg/ml liquid ☎ ▼ Hazardous drug handling precautions	No significant difference in efficacy for moderate-to-severe plaque psoriasis between MTX vs CSA in network meta-analysis.[62] Prefer intermittent, short-term (3-6mo) use rather than continuous long term use to ↓AE.	AE: HA, HTN, hypertrichosis, gingival hyperplasia. Serious: cytopenias, infection, nephrotoxicity, skin cancer, lymphoma M: Baseline: BP, CBC, SCr, eGFR, lipids, LFTs, Mg, K, uric acid, tuberculosis, pregnancy Ongoing: BP, CBC, SCr, eGFR, lipids, LFTs, electrolytes; rebound psoriasis with abrupt D/C.	1-5mg/kg/day divided q12h. (Max: 5mg/kg/d x 1-2yr) ↓ by 0.5-1mg/kg/day if adequate disease control, ↑BP, &/or ↓renal fx.	$200-600
Methotrexate (MTX), g 2.5mg, 10mg tablets 10, 25mg/mL vial (10, 12.5, 15, 17.5, 20, 22.5, 25mg)[injection] ♣ 50mg/mL syringe (7.5, 10, 12.5, 15, 17.5, 20, **22**.5, 25mg)♣ Subcut autoinjector kit 25mg/mL (7.5, 10, 15, 20, 25)	Can be given parenterally to ↓GI AE.[8] MTX can be used in combination with biologics and phototherapy.[63]	AE: mouth sores, GI, N, hair loss. {**Folic acid** 1-5mg/d po on non-MTX day to ↓ GI AEs} Severe: liver damage, bone marrow suppression, pulmonary fibrosis, cutaneous cancer, lymphoproliferative malignancy. M: Baseline: CBC, SCr, eGFR, LFT, hepatitis B/C, tuberculosis/HIV screen, pregnancy. Ongoing: CBC, LFTs q2-4wks & 1 wk after dose ↑, then q2wk x 1mo, then q2-3mo; SCr, eGFR q6-12mo; lung function.	Initial: 7.5-10mg q1wk then ↑q4-8wk PO/subcut/IM Usual: 10-25mg/wk (Max: 30mg/wk)	~<$25 po 10mg SC $75 25mg SC $95
Mycophenolate Mofetil (MMF) ☎ ▼ CELLCEPT, g 250mg cap, 500mg tab **Mycophenolate Sodium** ☎ ▼ MYFORTIC 180, 360mg EC tabs g AKA mycophenolic acid	AE: headache, GI (diarrhea, nausea, vomiting, abdominal cramps), ↑cholesterol, ↑BP Severe: cytopenia, infections, cutaneous cancer, lymphoproliferative malignancy.	M: Baseline: BP, CBC, SCr, eGFR, LFT, hepatitis B/C, tuberculosis/HIV screen, pregnancy. Ongoing: BP, CBC, LFT q2wk x 1mo, then q1mo x 3mo, then q2-3mo.	Initial: 1-2g/d po daily or divided BID (Max: 3g/d)	60-110 160
JANUS KINASE INHIBITORS (JAK-Θ):[64,65] Inhibition of JAK1 prevents signaling of IL-4, IL-13, & other cytokines involved in the pathogenesis of atopic dermatitis. **Tofacitinib** XELJANZ & **baricitinib** OLUMIANT are considered **off-label**. JAK-Θ are used where other systemic tx (including biologics) are contraindicated or ineffective. Refer to specialist. **Boxed warnings**: ↑ risk of serious infection, mortality, malignancies, MACE, & thrombosis. M: Baseline: CBC, lipids, LFT, viral hepatitis, TB, pregnancy, BG in DM, eGFR, SCr; ≥1 mo: CBC, LFT, eGFR, SCr; ≥3 mo followed by q3-6mo: CBC, LFTs, eGFR, SCr, lipids, BG in DM. **Immunizations:** Complete all age-appropriate immunizations before initiating therapy; avoid administration of **live vaccines** immediately prior to, during, and immediately after therapy. Vaccinate for shingles prior to starting JAK-Θ.				
Abrocitinib CIBINQO ☎ ✆ 50mg, 100mg, 200mg tablets	✓Moderate-severe atopic dermatitis, ≥12yr	AE: GI (N, pain, vomiting), nasopharyngitis, HA, acne, HSV, ↑CK, dizziness, ↑BP, ↑lipids, ↓BG in DM. Serious: infection (e.g. pneumonia), GI perforation, neutropenia, hepatotoxicity (rare) CI: antiplatelet use (except ASA ≤ 81mg/d) during first 3mo of tx[US FDA]. DI: Avoid antiplatelets-may ↑ antiplatelet effects of abrocitinib	100mg po daily, ↑ 200mg/d in 12wks if inadequate response (Max: 200mg/d) eGFR < 60mL/min: 50mg/d (Max: 100mg/d)	$1500 $1700
Deucravacitinib SOTYKTU 8 ✗ ⊗ 6mg tablet	✓Moderate-severe plaque psoriasis, ≥18yr {~4 months to see benefit}	AE: **Common:** URTI, ↑CK, HSV, mouth ulcers, folliculitis, acne. Other: malignancies (e.g. lymphoma), rhabdomyolysis, ↑LFT, ↑TG CI: Avoid in severe hepatic impairment (Child Pugh Class C)	6mg po daily	$1200
Upadacitinib RINVOQ ☎ ✆ 15mg, 30mg, 45mg XR tablets	✓Moderate-severe atopic dermatitis, ≥12yr	AE: URTI, acne, HA, peripheral edema, folliculitis, ↑CK, ↑lipids, ↑LFT, ↑wt. Serious: anemia (leuko-, lympho-, neutro-penia), infection (e.g. HSV, influenza, pneumonia)	15mg po daily (Max: 30mg/d) eGFR 15-30mL/min: 15mg/d eGFR < 15mL/min: not recommended	$1600 ($2400)

Icotrokinra ICOTYDE Oral IL-23 receptor antagonist for moderate-severe plaque psoriasis ≥ 12y; AE: HA, nausea, cough, fungal infection, fatigue; DI: Avoid live vaccines; 200mg po qam (water & empty stomach); $8285/30d; long-term efficacy and safety unknown.

Drug-induced psoriasis: beta blockers, biologics (e.g. adalimumab, dupilumab, infliximab), bupropion, hydroxychloroquine, lithium, terbinafine.[15-17] Drug-induced seborrheic dermatitis: auranofin, fluorouracil, griseofulvin, haloperidol, lithium, psoralen.[18]

Abbreviations: ☎ EDS in SK ✆=prior approval required by NIHB ✗ =non formulary SK ⊗-not covered by NIHB =Limit/avoid sun & UV light exposure due to increased risk of photosensitivity, phototoxicity, sunburn or skin cancer. Use appropriate sun protection.
AD=atopic dermatitis **AhR**=aryl hydrocarbon receptor **AZA**=azathioprine **BG**=blood glucose **BID**=twice daily **BMD**=bone mineral density **BSA**=body surface area **Ca++**=calcium **CK**=creatine kinase **CNI**=calcineurin inhibitor **crm**=cream **CT**=coal tar **CVD**=cardiovascular disease **D**=diarrhea **D/C**=discontinue **DM**=diabetes mellitus **DVT**=deep vein thrombosis **dx**=disease **eGFR**=estimated glomerular filtration rate **GI**=gastrointestinal **HA**=headache **HSV**=herpes simplex virus **lot**=lotion **LFT**=liver function test **JAK-Θ**=janus kinase inhibitor(s) **MACE**=major adverse cardiovascular events **MI**=myocardial infection **MMF**=mycophenolate mofetil **MTX**=methotrexate **N**=nausea **OC**=oral contraceptive **Oint**=ointment **PDE4-Θ**=phosphodiesterase-4 inhibitor(s) **PE**=pulmonary embolism **PIM**=pimecrolimus **PP**=plaque psoriasis **PTH**=parathyroid hormone **SA**=salicylic acid **SCr**=serum creatinine **SebD**=seborrheic dermatitis **SK**=Saskatchewan **TAC**=tacrolimus **TAZ**=tazarotene **TB**=tuberculosis **TCS**=topical corticosteroid(s) **TG**=triglycerides **TNF**=tumor necrosis factor **tx**=treatment **URTI**=upper respiratory tract infection **UTI**=urinary tract infection **UVA**=ultraviolet A **V**=vomiting **VDA**=vitamin D analogue **wk**=week

ATOPIC DERMATITIS (ECZEMA), PSORIASIS, & SEBORRHEIC DERMATITIS: Comparison Chart

Acknowledgements:
Originally written by: Margaret Jin (with original draft by Devanshi Parekh). **Revised/updated/ongoing edits by**: Margaret Jin (2026-)
Disclosures: No conflicts of interest are reported
Thanks to reviewers of the 16th Edition (2027): Loren Regier, Amy Wiebe, Brent Jensen
Thanks to our previous reviewers of this chart:

Copyright 1997 RxFiles Academic Detailing, University of Saskatchewan. All Rights Reserved. Use of this work, in whole or in part, for the purpose of developing, improving, testing, training, or otherwise operating any artificial intelligence, machine learning, or automated decision-making system is not permitted by the copyright holder. This includes but is not limited to using a portion of the work as a data input for the purpose of querying an existing third-party AI system. No part of this work may be reproduced, distributed, or transmitted in any form or by any means, including photocopying, recording, or other electronic or mechanical methods, without the prior written permission of the copyright holder. For copyright permission requests, please contact **info@RxFiles.ca**.

Disclaimer: RxFiles Academic Detailing is part of the College of Pharmacy and Nutrition at the University of Saskatchewan. The content of this work represents the research, experience and opinions of the authors and not those of the University of Saskatchewan. Neither the authors nor the University of Saskatchewan nor any other party who has been involved in the preparation or publication of this work warrants or represents that the information contained herein is accurate or complete, and they are not responsible for any errors or omissions or for the result obtained from the use of such information. Any use of the materials will imply acknowledgment of this disclaimer and release any responsibility of the University of Saskatchewan, its employees, servants or agents. Readers are encouraged to confirm the information contained herein with other sources.

💻 Online Extras

Atopic dermatitis

- JAK-inhibitors– abrocitinib, baricitinib, upadacitinib
- Biologics: dupilumab, lebrikizumab, nemolizumab, tralokinumab ☎. - ☎ fail/intolerant to topical + phototherapy[if available] & one of MTX, cyclosporine, MMF, or AZA

Biologics for plaque psoriasis:

- TNF-Alpha-Ɵ: etanercept, adalimumab, infliximab, certolizumab
- IL12/23-Ɵ – ustekinumab
- IL17A&IL17F-Ɵ – bimekizumab
- IL-17A-Ɵ – brodalumab, ixekizumab, secukinumab
- IL23-Ɵ – guselkumab, risankizumab, tildrakizumab

Search Terms

Aloe Vera	21
ALPHA KERI	21
Atopic Dermatitis	21
AVEENO	21
CERAVE	21
CETAPHIL	21
Ciclopirox	21
Clobetasol	21
CLOBEX	21
Coal Tar	21
Corticosteroid	21
DERMAL THERAPY	21
DERMAREST	21
Dermatitis	21
DERMOVATE	21
DORMER	21
Eczema	21
EUCERIN	21
GARNIER PURIFYING	21
GLAXAL BASE	21
GLYCERIN	21
GLYSOMED	21
Hydrocortisone	21
HYDROVAL	21
Ketoconazole	21
KETODERM	21
LOPROX	21
LUBRIDERM	21
Moisturizer	21
NEOSTRATA	21
NIZORAL	21

NUTRADERM	21
NUTROGENA	21
Psoriasis	21
Salicylic Acid	21
SEBCUR	21
Seborrhea	21
SELSUN BLUE	21
Steroid	21
TARDAN	21
Topical Corticosteroids	21
UREMOL	21
URISEC	21
VASELINE	21
Anthralin	22
ANZUPGO	22
ARAZLO	22
Atopic Dermatitis	22
Betamethasone Dipropionate	22
Calcipotriol	22
Calcitriol	22
Crisaborole	22
Delgocitinib	22
Dermatitis	22
DIPROSALIC	22
DOVONEX	22
DUOBRII	22
Eczema	22
ELIDEL	22
ENSTILAR	22
EURCRISA	22
Halobetasol	22
NDUVRA	22
Non-Corticosteroid	22
OPZELURA	22
Pimecrolimus	22
PROTOPIC	22
Psoriasis	22
Roflumilast	22
Ruxolitinib	22
Salicylic Acid	22
Seborrhea	22
SILKIS	22
Tacrolimus	22
Tapinarof	22
Tazarotene	22
VTAMA	22
WYNZORA	22
ZORYVE	22
Abrocitinib	23
Acitretin	23
Apremilast	23
CELLCEPT	23
CIBINQO	23
Cyclosporine	23
Deucravacitinib	23
Icotrokinra	23
ICOTYDE	23
Methotrexate	23
Mycophenolate	23
MYFORTIC	23
NEORAL	23
OTELZA	23
Psoriasis	23
RINVOQ	23
SORIATANE	23
SOTYKTU	23
Upadacitinib	23

References for TOPICAL NON-CORTICOSTEROID FOR ATOPIC DERMATITIS (ECZEMA), PSORIASIS, & SEBORRHEIC DERMATITIS: Comparison Chart

1. Pharmacist's Letter. Use a stepwise approach when treating eczema. Pharmacist's Letter. August 2025. Accessed March 20, 2026. https://pharmacist.therapeuticresearch.com
2. BMJ Best Practice. Eczema (atopic dermatitis). BMJ Publishing Group. Topic last updated October 23, 2025. Accessed March 20, 2026. https://bestpractice.bmj.com/topics/en-gb/87 [bestpractice.bmj.com]
3. Davis DMR, Alikhan A, Bercovitch L, et al. Guidelines of care for the management of atopic dermatitis in pediatric patients. J Am Acad Dermatol. Published online April 28, 2026. doi:10.1016/j.jaad.2026.02.113.
4. Drucker A. Atopic dermatitis. Ann Intern Med. 2026;179(3):ITC33-ITC48. doi:10.7326/ANNALS-25-05580.
5. Yadav G, Miller-Monthrope Y, Rao J, et al. Optimizing the management of psoriasis in patients with skin of color: A Canadian Delphi consensus. JAAD Int. 2024;19:12-20. Published November 14, 2024. doi:10.1016/j.jdin.2024.09.015.
6. Stein Gold L, Alexis A, Glick BP, et al. Management of challenging psoriasis clinical scenarios. Dermatol Ther (Heidelb). 2025;15(6):1555-1567. doi:10.1007/s13555-025-01393-3.
7. Nicholas MN, Chan AR, Hessami-Booshehri M. Psoriasis in patients of color: Differences in morphology, clinical presentation, and treatment. Cutis. 2020;106(2S):7-e10. doi:10.12788/cutis.0038.
8. LeFevre NM, Braudis K, Feigenbaum LS. Seborrheic dermatitis: Diagnosis and treatment. Am Fam Physician. 2025;112(2):166-173.
9. Gupta S, Ramam M, Sharma VK, Sethuraman G, Pandey RM, Bhari N. Evaluation of a paraffin-based moisturizer compared with a ceramide-based moisturizer in children with atopic dermatitis: A double-blind randomized controlled trial. Pediatr Dermatol. 2023;40(4):627-632. doi:10.1111/pde.15355.
10. Wei J, et al. Alcohol consumption and smoking in relation to psoriasis: A Mendelian randomization study. Br J Dermatol. 2022;187(5):684-691.
11. Gamret AC, et al. Complementary and alternative medicine therapies for psoriasis: A systematic review. JAMA Dermatol. 2018;154(11):1330-1337.
12. AAAAI/ACAAI Joint Task Force on Practice Parameters; Chu DK, Schneider L, et al. Atopic dermatitis (eczema) guidelines: 2023 practice parameters. Ann Allergy Asthma Immunol. 2024;132(3):274-312. doi:10.1016/j.anai.2023.11.009.
13. Sidbury R, Alikhan A, Bercovitch L, et al. Guidelines of care for the management of atopic dermatitis in adults with topical therapies. J Am Acad Dermatol. 2023;89(1):e1-e20. doi:10.1016/j.jaad.2022.12.029.
14. Apfelbacher CJ, van Zuuren EJ, Fedorowicz Z, Jupiter A, Matterne U, Weisshaar E. Oral H1 antihistamines as monotherapy for eczema. Cochrane Database Syst Rev. 2013;(2):CD007770. doi:10.1002/14651858.CD007770.pub2.
15. Balak DM, Hajdarbegovic E. Drug-induced psoriasis: clinical perspectives. Psoriasis (Auckl). 2017;7:87-94. Published 2017 Dec 7. doi:10.2147/PTT.S126727
16. Balestri R, Magnano M, Infusino SD, et al. Unravelling the Mystery of Psoriasis Dermatitis (PsoDermatitis): A Practical Guide to Recognition and Management. J Clin Med. 2025;15(1):130. Published 2025 Dec 24. doi:10.3390/jcm15010130
17. Qian SY, Goyal J, Wang S, Vender R. Drug-Induced Generalized Pustular Psoriasis: A Systematic Review. J Cutan Med Surg. Published online August 11, 2025. doi:10.1177/12034754251364849l8. American Academy of Dermatology. Seborrheic dermatitis: causes. American Academy of Dermatology Association. Accessed March 20, 2026. https://www.aad.org/public/diseases/a-z/seborrheic-dermatitis-causes.
19. Lio P. Ask the Doctor: Alternative therapies for treating eczema. Eczema Society of Canada. https://eczemahelp.ca/ask-the-doctor-alternative-therapies-for-treating-eczema/ accessed June 27, 2026.
20. Elmets CA, Korman NJ, Prater EF, et al. Joint AAD–NPF guidelines of care for the management and treatment of psoriasis with topical therapy and alternative medicine modalities. J Am Acad Dermatol. 2021;84(2):432-470. doi:10.1016/j.jaad.2020.07.087.
21. Roelofzen JH, Aben KK, Oldenhof UT, et al. No increased risk of cancer after coal tar treatment in patients with psoriasis or eczema. J Invest Dermatol. 2010;130(4):953-961.
22. Roelofzen JHJ, Aben KKH, van de Kerkhof PCM, van der Valk PGM, Kiemeney L. Dermatological exposure to coal tar and bladder cancer risk: A case-control study. Urol Oncol. 2015;33(1):20.e19-20.e22.
23. Davis DMR, Frazer-Green L, Alikhan A, et al. Focused update: Guidelines of care for the management of atopic dermatitis in adults. J Am Acad Dermatol. 2025;93(3):745.e1-745.e7. doi:10.1016/j.jaad.2025.05.1386.
24. Yin Z, Xu J, Luo D. Efficacy and tolerance of tacrolimus and pimecrolimus for atopic dermatitis: a meta-analysis. J Biomed Res. 2011;25(6):385-391.
25. El-Batawy MM, Bosseila MA, Mashaly HM, Hafez VS. Topical calcineurin inhibitors in atopic dermatitis: a systematic review and meta-analysis. J Dermatol Sci. 2009;54(2):76-87.
26. US Food and Drug Administration. FDA issues public health advisory informing health care providers of safety concerns associated with the use of two eczema drugs, Elidel and Protopic. March 10, 2005. Accessed March 20, 2026. https://www.fda.gov/bbs/topics/ANSWERS/2005/ANS01343.html.
27. Health Canada. Health Canada advisory: Elidel (pimecrolimus) and Protopic (tacrolimus). April 2005. Accessed March 20, 2026. https://www.hc-sc.gc.ca/english/protection/warnings/2005/2005_31.html.
28. Canadian Dermatology Association. Topical calcineurin inhibitors position statement. May 2018. Accessed March 20, 2026. https://dermatology.ca/tci-position-statement/.
29. Margolis DJ, Abuabara K, Hoffstad OJ, et al. Association between malignancy and topical use of pimecrolimus. JAMA Dermatol. 2015;151(6):594-599.
30. Chu A, Devasenapathy N, Wong M, et al. Cancer risk with topical pimecrolimus and tacrolimus for atopic dermatitis: systematic review and Bayesian meta-analysis. Br J Dermatol. 2023;188(Suppl 2):ljac140.017. doi:10.1093/bjd/ljac140.017
31. Asgari MM, Tsai AL, Avalos L, Sokil M, Quesenberry CP. Association between topical calcineurin inhibitor use and keratinocyte carcinoma risk among adults with atopic dermatitis. JAMA Dermatol. 2020;156(5):548-555.
32. He Y, Liu J, Wang Y, et al. Topical administration of crisaborole in mild to moderate atopic dermatitis: a systematic review and meta-analysis. Dermatol Ther. 2023;36:e1869934. doi:10.1155/2023/1869934.
33. Devasenapathy N, Chu A, Wong M, et al; AAAAI/ACAAI Joint Task Force on Practice Parameters for Atopic Dermatitis Guideline Development Group. Cancer risk with topical calcineurin inhibitors, pimecrolimus and tacrolimus, for atopic dermatitis: a systematic review and meta-analysis. Lancet Child Adolesc Health. 2023;7(1):13-25. doi:10.1016/S2352-4642(22)00283-8.
34. Fahrbach K, Tarpey J, Washington EB, et al. Crisaborole ointment, 2%, for treatment of patients with mild-to-moderate atopic dermatitis: systematic literature review and network meta-analysis. Dermatol Ther (Heidelb). 2020;10(4):681-694. doi:10.1007/s13555-020-00389-5.
35. Pharmacist's Letter Canada. Don't jump to topical Zoryve for psoriasis. Pharmacist's Letter Canada. July 2023. Accessed March 20, 2026. https://ca-pharmacist.therapeuticresearch.com/Content/Articles/PLC/2023/Jul/Don-t-Jump-to-Topical-Zoryve-for-Psoriasis.
36. Aleid AM, Mohammed Alshreef S, Tariq Alrubaiaan M, et al. Effectiveness and safety of roflumilast in the treatment of psoriasis: a systematic review and meta-analysis. Curr Drug Saf. Published online February 10, 2026. doi:10.2174/0115748863406187251031101244.
37. Kow CS, Ramachandram DS, Hasan SS, Thiruchelvam K. Efficacy and safety of roflumilast cream in atopic dermatitis: a systematic review and meta-analysis of randomized controlled trials. Australas J Dermatol. 2025;66(4):220-224. doi:10.1111/ajd.14466.

38. Lebwohl M, Ast E, Callen JP, et al. Once-daily tazarotene gel versus twice-daily fluocinonide cream in the treatment of plaque psoriasis. J Am Acad Dermatol. 1998;38(5 Pt 1):705-711. doi:10.1016/S0190-9622(98)70594-8.
39. The Medical Letter on Drugs and Therapeutics. Comparison table: some drugs for plaque psoriasis. The Medical Letter on Drugs and Therapeutics Adapted for Canada. September 30, 2024.
40. Ashcroft DM, Po AL, Williams HC, Griffiths CE. Systematic review of comparative efficacy and tolerability of calcipotriol in treating chronic plaque psoriasis. BMJ. 2000;320(7240):963-967. doi:10.1136/bmj.320.7240.963.
41. Ortonne JP, Humbert P, Nicolas JF, et al. Intra-individual comparison of the cutaneous safety and efficacy of calcitriol 3 µg/g ointment and calcipotriol 50 µg/g ointment on chronic plaque psoriasis localized in facial, hairline, retroauricular, or flexural areas. Br J Dermatol. 2003;148(2):326-333. doi:10.1046/j.1365-2133.2003.05228.x
42. Körver JE, Vissers WH, van Rens DW, et al. A double-blind, randomized quantitative comparison of calcitriol ointment and calcipotriol ointment on epidermal cell populations, proliferation, and differentiation. Br J Dermatol. 2007;156(1):130-137. doi:10.1111/j.1365-2133.2006.07561.x
43. Franssen ME, de Jongh GJ, van Erp PE, van de Kerkhof PC. A left/right comparison of twice-daily calcipotriol ointment and calcitriol ointment in patients with psoriasis: the effect on keratinocyte subpopulations. Acta Derm Venereol. 2004;84(3):195-200. doi:10.1080/00015550410025930.
44. Bourke JF, Iqbal SJ, Hutchinson PE. A randomized double-blind comparison of the effects on systemic calcium homeostasis of topical calcitriol (3 µg/g) and calcipotriol (50 µg/g) in the treatment of chronic plaque psoriasis vulgaris. Acta Derm Venereol. 1997;77(3):228-230. doi:10.2340/0001555577228230.
45. Guenther LC, Poulin YP, Pariser DM. A comparison of tazarotene 0.1% gel once daily plus mometasone furoate 0.1% cream once daily versus calcipotriene 0.005% ointment twice daily in the treatment of plaque psoriasis. Clin Ther. 2000;22(10):1225-1238. doi:10.1016/S0149-2918(00)83065-9.
46. Kragballe K, Hoffmann V, Ortonne JP, Tan J, Nordin P, Segaert S. Efficacy and safety of calcipotriol plus betamethasone dipropionate scalp formulation compared with calcipotriol scalp solution in the treatment of scalp psoriasis: a randomized controlled trial. Br J Dermatol. 2009;161(1):159-166. doi:10.1111/j.1365-2133.2009.09116.x
47. van de Kerkhof PC, Hoffmann V, Anstey A, Barnes L, Bolduc C, Reich K, Saari S, Segaert S, Vaillant L. A new scalp formulation of calcipotriol plus betamethasone dipropionate compared with each of its active ingredients in the same vehicle for the treatment of scalp psoriasis: a randomized, double-blind, controlled trial. Br J Dermatol. 2009;160(1):170-176. doi:10.1111/j.1365-2133.2008.08927.x
48. Mason AR, Mason J, Cork M, Dooley G, Hancock H. Topical treatments for chronic plaque psoriasis. Cochrane Database Syst Rev. 2013;(3):CD005028. doi:10.1002/14651858.CD005028.pub3
49. Armstrong AW, Read C. Pathophysiology, clinical presentation, and treatment of psoriasis: a review. JAMA. 2020;323(19):1945-1960. doi:10.1001/jama.2020.4006.
50. Kircik LH, Papp KA, Stein Gold L, Harris S, Pharm TL, Pillai R. Assessing the synergistic effect of a fixed combination halobetasol propionate 0.01% and tazarotene 0.045% lotion in moderate-to-severe plaque psoriasis. J Drugs Dermatol. 2019;18(3):279-284.
51. Ramachandran V, Bertus B, Bashyam AM, Feldman SR. Treating psoriasis with halobetasol propionate and tazarotene combination: a review of phase II and III clinical trials. Ann Pharmacother. 2020;54(9):872-878. doi:10.1177/1060028020910439.
52. Del Rosso JQ, Kircik L, Lin T, Pillai R. Halobetasol 0.01%/Tazarotene 0.045% Fixed-combination Lotion in the Treatment of Plaque Psoriasis: Sensitization and Irritation Potential. J Clin Aesthet Dermatol. 2019;12(1):11-15.
53. Purdy K, Gooderham M, Abu-Hilal M, et al. Supporting evidence and recommendations for the management of patients with systemic-eligible atopic dermatitis: A Canadian consensus document. *JAAD Int*. 2025;24:31-49. Published 2025 Sep 26. doi:10.1016/j.jdin.2025.06.009.
54. Gooderham MJ, Hong HC, Lynde C, et al. Canadian Consensus Guidelines for the Management of Atopic Dermatitis with Topical Therapies. *Dermatol Ther (Heidelb)*. 2025;15(6):1467-1485. doi:10.1007/s13555-025-01386-2.
55. Sidbury R, Alikhan A, Bercovitch L, et al. Guidelines of care for the primary prevention of atopic dermatitis and awareness of comorbid conditions in pediatric atopic dermatitis. *J Am Acad Dermatol*. 2026;95(1):124.e1-124.e19. doi:10.1016/j.jaad.2026.02.114
56. Davis DMR, Frazer-Green L, Alikhan A, et al. Focused update: Guidelines of care for the management of atopic dermatitis in adults. *J Am Acad Dermatol*. 2025;93(3):745.e1-745.e7. doi:10.1016/j.jaad.2025.05.1386
57. Chu AWL, Wen A, Guyatt GH, et al. Antihistamines for atopic dermatitis (eczema): systematic review and network meta-analysis of randomised trials. *BMJ*. 2026;394:e100163. Published 2026 Jul 29. doi:10.1136/bmj-2026-100163